



# Studying the spectral diffusion dynamics of chlorophyll *a* and chlorophyll *b* using two-dimensional electronic spectroscopy

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## ABSTRACT

Chlorophyll *a* (Chl *a*) and Chlorophyll *b* (Chl *b*) are major pigments in the photosynthetic machinery in green plants. Despite their similarities in chemical structures, the relaxation dynamics as observed in the ultrafast methods, such as pump probe and multidimensional spectroscopy, demonstrate a distinct kinetics. Here, we employ the ultrafast two-dimensional electronic spectroscopy (2DES) method to characterize the frequency fluctuation correlation function (FFCF) for Chl *a* and Chl *b* in various solvent environments, through the Centre Line Slope (CLS) method. We observe that the FFCF decay for both Chl *a* and Chl *b* in the methanol solvent environment has a ~40 ps component that is absent in aprotic solvents, tetrahydrofuran and diethyl ether. We observe also that Chl *b* exhibit higher values of inhomogeneous broadening than Chl *a*.

## 1. Introduction

Chlorophyll *a* (Chl *a*) and Chlorophyll *b* (Chl *b*) are the basic building blocks of natural light-harvesting complexes (LHC) which occur in plants and green algae. They have been extensively studied due to their widespread occurrence and important functional roles in the LHCs. Essentially, the chlorophylls (Chls) in LHCs capture light energy and deliver it to the reaction center through a highly optimized energy transfer network on a femtosecond to picosecond timescales [1–5]. The optimization of this process is governed by the organization of Chls within the protein matrix [6–9]. It is well established that Chls are organized in a complex and orderly manner within the lipoprotein through a specific (hydrogen bond, donor-acceptor complex formation) and non-specific interaction (hydrophobic and dispersive forces) [10–14]. Spectroscopic studies on the isolated systems of Chls in various solvents have been widely studied to identify this interaction and ultimately understand the role of local protein environment in light-harvesting complexes. A review by Muring *et al.* summarizes the Chl *a* absorption and emission spectra from > 50 solvents belonging to different groups (protic, non-polar, aromatic, etc.) [15]. It describes the mechanisms of solvatochromism and differentiation between the specific and non-specific interactions in Chl *a*. However, studies on steady-state measurement do not explain the details on time-dependent

processes such as the solvent reorganization mechanism and spectral modulation which commonly occurs on the sub-picosecond to hundreds of picosecond timescales. To uncover such information, ultrafast nonlinear spectroscopic techniques such as time-resolved fluorescence (TRF) [16] and three-pulse photon echo peak shift (3PEPS) [17] has been applied. Two-dimensional electronic spectroscopy (2DES) is a nonlinear optical spectroscopic tool that has become a powerful technique to elucidate the dynamics in condensed matter [18–21]. In recent studies, 2DES was employed to characterize the spectral diffusion dynamics of Chl *a* in a whole broad range of solvent environments [22–24]. The results indicate the amplitudes and time scales of frequency fluctuation and were attributed to the solvent reorganization effects occurring after the laser excitation. However, little has been reported on Chl *b*, specifically, the spectral diffusion dynamics of Chl *b* monomer.

Despite their close resemblance in structure (Fig. 1), the spectral properties of Chl *a* and Chl *b* molecules are quite distinct. For instance, in the linear spectrum (Fig. 2), the 1st excited state ( $Q_y$  band) of Chl *b* can be seen to be blue shifted with respect to the Chl *a* [25,26]. Apart from that, studies on the ultrafast relaxation dynamics, such as transient absorption and multidimensional spectroscopy, demonstrate distinct kinetics between the two molecules. It was shown that the internal conversion between the  $Q_x$  (the 2nd excited state of Chls) and  $Q_y$  bands

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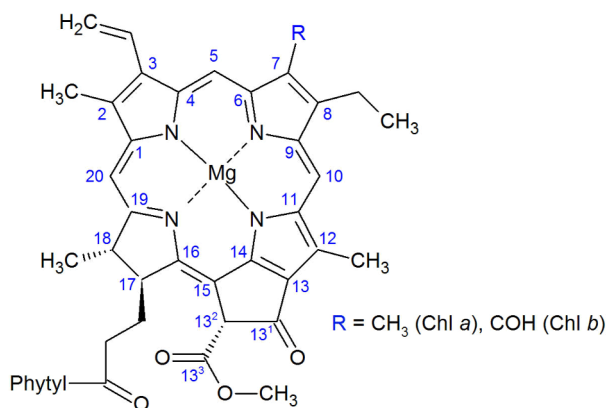


Fig. 1. Structure of Chlorophyll a and Chlorophyll b with IUPAC numbering.

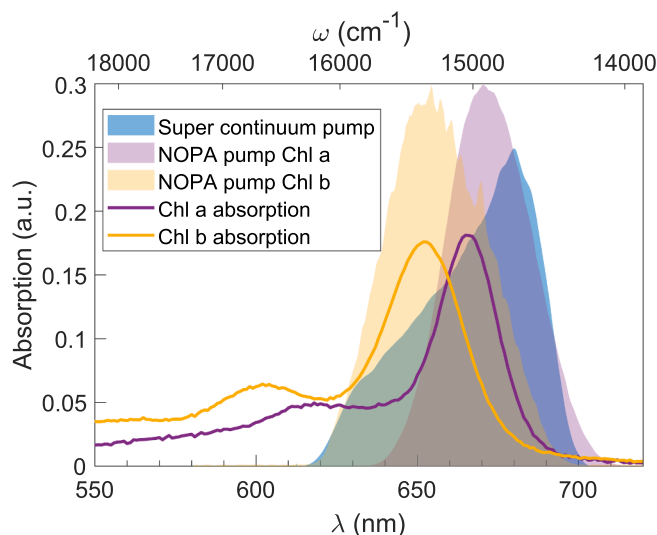


Fig. 2. Representative linear absorption spectra of Chl a (purple line) and Chl b (yellow line) in MeOH. The shaded overlays depict excitation pulse spectra.

is faster in Chl a compared to Chl b [27]. This was attributed to the stronger mixing between the  $Q_x$  and  $Q_y$  bands in Chl a [28]. More recently, a new technique called the two-dimensional electronic-vibrational spectroscopy (2DEV) developed by Fleming *et al.* has revealed the specific molecular vibrational spectra assigned to each type of the Chl allowing further distinction of these molecules [29].

In this article, we present results from the 2DES of monomer Chl a and Chl b molecules in different solvents and investigate the effect of specific interactions that occurred in Chl-solvent systems. We applied the center line slope method ( $\text{CLS}\omega_t$ ) devised by Fayer *et al.* [30] to retrieve the normalized frequency fluctuation correlation function (FFCF) from the 2D spectra. This method has been mostly used to study spectral diffusion in the 2DIR regime, and it is sensitive towards the solute-solvent interaction. Here, we measure and discuss the spectral diffusion dynamics as measured from the obtained CLS values for Chl a and Chl b in various solvents. We observed distinct  $\text{CLS}\omega_t$  values for each of Chls in different solvents. We discuss these findings in relation to the modification at the periphery of the Chl molecules which influences the structural fluctuation and thus the frequency modulation of the excited state transition.

## 2. Materials and methods

Chl a and Chl b were purchased from Sigma-Aldrich and used without further purification. All the samples were dissolved in various HPLC and spectroscopic grade solvents (listed in Table 1) to match an

Table 1

Physical properties of used solvents and parameters of the  $Q_y$  absorption peaks of Chls a and b. Where  $\omega_{FWHM}$  represents the full-width at half-maximum of the  $Q_y$  absorption peaks.

| Solvent           | Viscosity <sup>a</sup> (cP)<br>at 298.15 K | Polarity <sup>b,c</sup> $E_T^N$<br>/Dielectric<br>constant, $\epsilon_r$ <sup>b</sup> | Chl a $\omega_{Max}$<br>[ $\omega_{FWHM}$ ]<br>( $\text{cm}^{-1}$ ) | Chl b $\omega_{Max}$<br>[ $\omega_{FWHM}$ ]<br>( $\text{cm}^{-1}$ ) |
|-------------------|--------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|
| MeOH              | 0.53                                       | 0.76/32.66                                                                            | 15,040 [5 2 0]                                                      | 15,380 [7 5 0]                                                      |
| THF               | 0.461                                      | 0.21/7.58                                                                             | 15,040 [3 9 0]                                                      | 15,400 [4 0 0]                                                      |
| Et <sub>2</sub> O | 0.235                                      | 0.12/4.20                                                                             | 15,110 [3 9 0]                                                      | 15,520 [4 0 0]                                                      |

<sup>a</sup> Obtained from Ref. [34].

<sup>b</sup> Obtained from Ref. [35].

<sup>c</sup> Water  $E_T^N = 1.00$ .

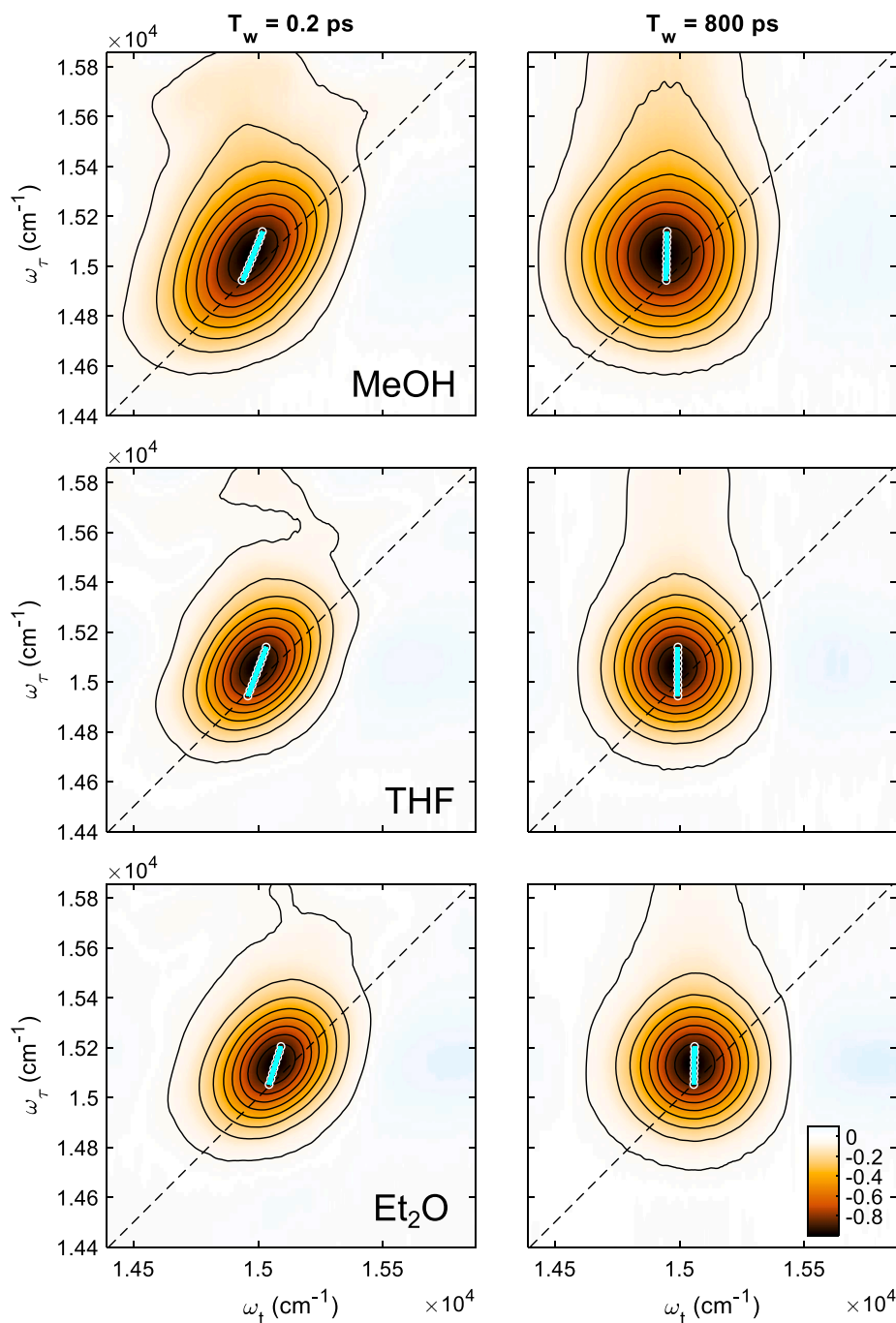
optical density (OD) of  $\sim 0.2$  at the  $Q_y$  band. As the oscillator strength of  $Q_y$  transition of Chl b is  $\sim 30\%$  lower than Chl a, to match the OD of 0.2, the concentration of the Chl b sample solution is  $\sim 1.5$  times higher than Chl a's. After the sample was prepared, it was used immediately to collect 2D spectra at room temperature (295 K). The sample degradation is rigorously tested and ensured to be  $< 5\%$  change by recording the linear absorption spectra and thin layer chromatograms before and after every 2D measurements. 2D spectra were collected for both Chl a and Chl b in protic solvent methanol (MeOH), and solvents with lower polarity: tetrahydrofuran (THF) and diethyl ether (Et<sub>2</sub>O).

The experimental setup for 2DES measurement using a pump-probe geometry has been described and demonstrated previously [32]. We performed our experiments using two sources of laser pulses, both pumped using a commercial Ti: Sapphire Amplifier (Legend Amplifier, Coherence, 800 nm at 1 kHz repetition rate). The first source is a Non-Collinear Optical Parametric Amplifier (TOPAS White) with a minor portion of 800 nm light focused into a 2-mm Sapphire crystal to generate a white light continuum to serve as a probe pulse. The second source is based on gas filamentation by the 800 nm pulse focused using a 1-m focal length concave mirror in an Ar gas-filled hollow tube at pressure of 2.5 bar [33]. The generated supercontinuum pulse is then split by a 90:10 beamsplitter to serve as the pump and the probe pulses. All the excitation pulses (pump) from both setups were compressed using the AOPDF (Dazzler, Fastlite) pulse shaper and measured to be sub 50 fs. The probe pulse is compressed using a pair of chirped mirrors (DCM 9, Laser Quantum). The polarization of the probe light is set at the magic angle ( $54.7^\circ$ ) with respect to the pump polarization. The experiments were performed using both sources, and the data were found to be very similar and repeatable. In general, the stability of the excitation pulses from the super continuum is better and the data acquisition time for 2DES measurements can be reduced.

The ultrafast two-dimensional electronic spectroscopy (2DES) measurements were carried out using a partially collinear (pump-probe) geometry implementing a  $1 \times 2$  phase cycling scheme [36]. The first coherence time,  $\tau$ , was scanned using the AOPDF pulse shaper to delay the first two excitation pulses from 0 fs to 141 fs in intervals of 3 fs and attenuated to 40 nJ sum energy for the two pulses. Additionally, the experiment was done in a partially rotating frame by setting  $\omega_{ref}$  at  $13990 \text{ cm}^{-1}$  (715 nm). Pump and probe beams were spatially overlapped into a 1 mm flow cell after which the probe signal was directed to the spectrometer (HORIBA Jobin Yvon, TRIAX 190) equipped with a CCD detector (Princeton Instruments, Pixis 100) to record the 2D spectra. The population time ( $T_w$ ) was recorded to monitor the dynamics of spectral diffusion from  $-0.3$  ps to 800 ps. However, the data were only analysed after  $T_w \geq 100$  fs due to the presence of coherent artifact that spectrally overlapped with the signal.

## 3. Results and discussion

Fig. 2 shows the linear absorption spectra of Chl a and Chl b dissolved in MeOH superimposed with the spectra of representative pump



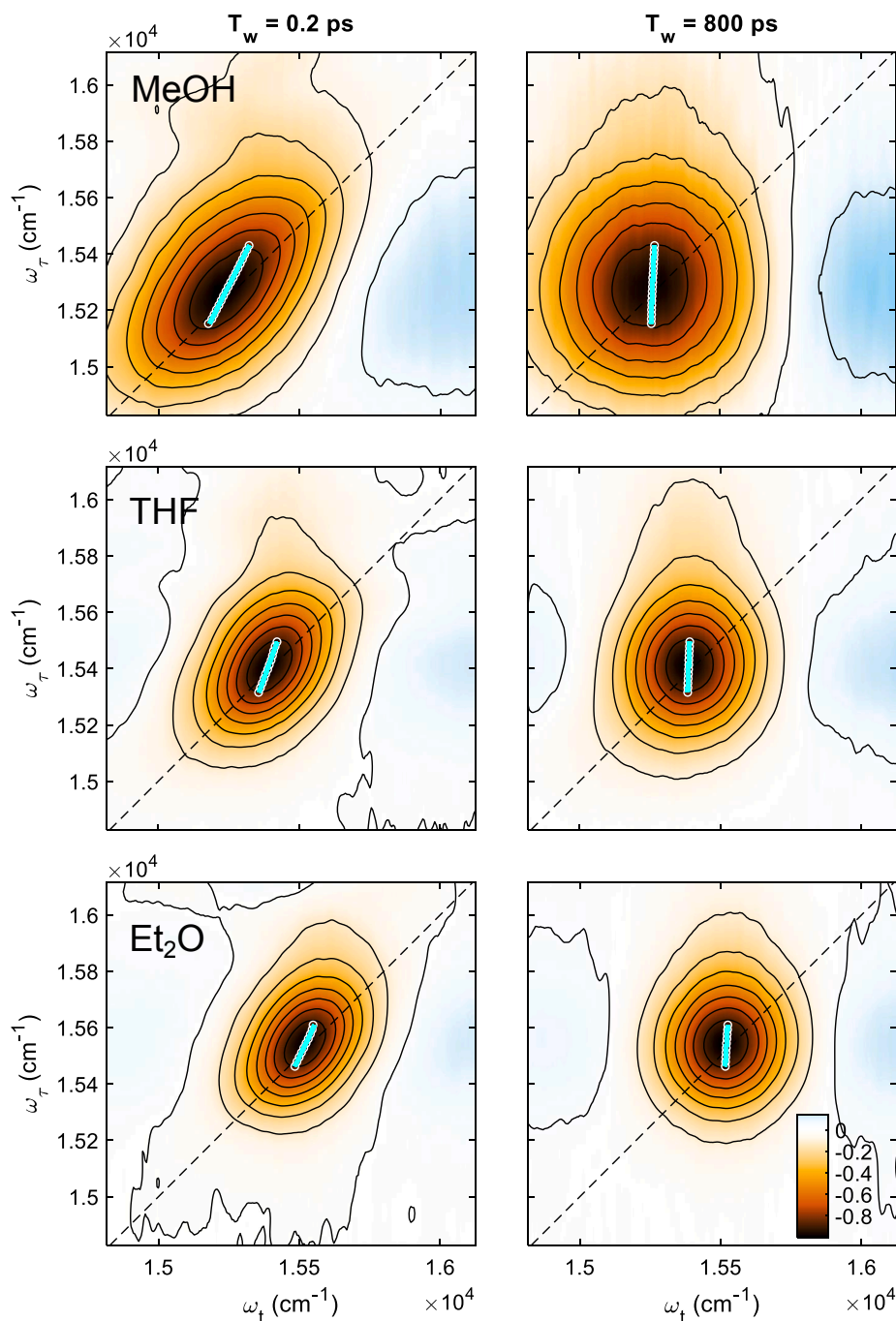
**Fig. 3.** 2D electronic spectra of Chl *a* in MeOH, THF and Et<sub>2</sub>O at waiting time  $T_w$  of 200 fs (left) and 800 ps (right) with corresponding center point (white circles) and center line fits overlaid (cyan line). The dashed line indicates diagonal line  $\omega_\tau$ (excitation axis) =  $\omega_t$ (detection axis). All spectra are normalized to the amplitudes of the minima.

excitations used for 2DES experiment. In all experiments, the laser spectrum was set to cover the  $Q_y$  excitation band.

Representative purely absorptive 2D spectra of Chl *a* and Chl *b* at two different delay times  $T_w$  were plotted in Figs. 3 and 4 respectively.

The response after pumping the  $Q_y$  band of Chl *a* and Chl *b* shows a strong negative signal which arises from ground state bleaching and stimulated emission of the 1st excited state. However, the signal strength in the 2D spectra for Chl *b* is lower than Chl *a* despite the same excitation energy was being used in the experiment. This is due to the lower oscillator strength of  $Q_y$  band in Chl *b* ( $f(Q_y) = 0.19$  in MeOH) compared to Chl *a* ( $f(Q_y) = 0.30$  in MeOH) [37]. As the 2DES signal scales as fourth order with respect to the transition dipole moments,

this results in substantially lower signal-to-noise ratio for the 2DES measurements of Chl *b* compared to Chl *a* under similar experimental conditions. This therefore normally require longer data acquisition time. As Chl *b* is more reactive compared to Chl *a*, this leads to a concern of longer exposure time under laser radiation for Chl *b*. Another feature difference between Chl *a* and Chl *b* that may have arose from the lower oscillator strength of  $Q_y$  band in Chl *b* is the more prominent positive band across  $\omega_t$  that is due to the excited-state absorption (ESA). Assuming the ESA to be similar for both Chl *a* and Chl *b*, the relatively higher oscillator strength of the  $Q_y$  band in Chl *a* may have caused the ESA band for Chl *a* to appear much less prominent than that of Chl *b*. The ESA band may also have caused the difference between Chl *a* and



**Fig. 4.** 2D electronic spectra of Chl *b* in MeOH, THF and Et<sub>2</sub>O at waiting time  $T_w$  of 200 fs (left) and 800 ps (right) with corresponding center point (white circles) and center line fits overlaid (cyan line). All notations are similar as Fig. 3.

Chl *b* in the shift of the peaks off the diagonal, normally assigned to dynamic Stokes shift. This uncalibrated effect from the ESA thus make any Stokes shift analysis here problematic.

In terms of the peak shape, a typical 2D spectra shows a bleach lobe that is initially elongated along the diagonal and gradually becomes symmetrical as  $T_w$  increases as can be seen in Figs. 3 and 4. This evolution is ascribed to the spectral diffusion process which causes an initially correlated spectrum to become uncorrelated at later  $T_w$  [38]. Quantification of these dynamics using CLS $\omega_\tau$  for Chl *a* dissolved in MeOH has been reported by Wells *et al.* [22]. More recently, Moca *et al.* reported the spectral diffusion dynamics of Chl *a* in various solvents using the ellipticity method. They concluded that the inhomogeneous broadening was promoted the most by viscosity and hydrogen bonding

solvents, and to a lesser degree by polarity [24]. Here, we use the CLS $\omega_t$  method to study the spectral diffusion dynamics of Chl *a* and Chl *b*. This method was chosen due to its robustness in retrieving the normalized FFCF [23,39,40]. CLS $\omega_t$  was carried out by using a Gaussian function fitted to a slice parallel to the detection axis  $\omega_t$  in a 2D spectra to determine the maximum signal position for the slice, which is the so-called center points. This was repeated for a series of slices at different excitation axis,  $\omega_\tau$ . In our analysis, we have analysed  $\omega_t$  in a range of 300 cm<sup>-1</sup> around the center of the signal to reduce the effect of noise at the sides of the peak. Collected in such way, series of center points were then fitted by linear regression to determine the center line, from which the gradient with respect to the vertical axis of the retrieved center line is defined as CLS $\omega_t$ . The fitted CLS $\omega_t$  from the 2D spectra is denoted by



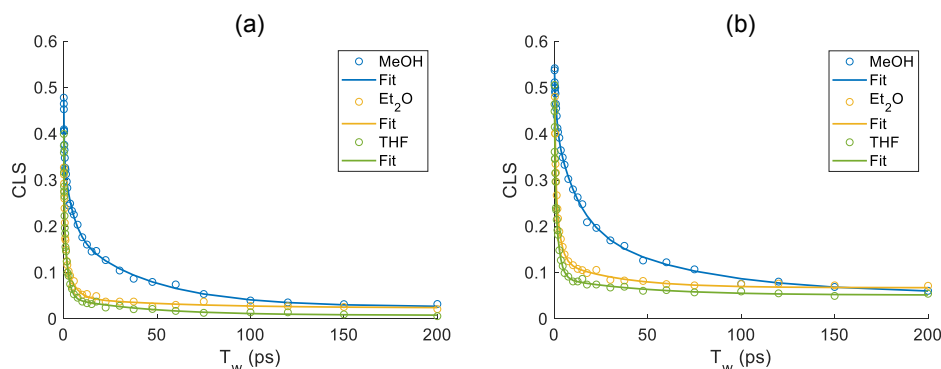


Fig. 5.  $\text{CLS}\omega_t$  vs  $T_w$  dynamics (circles) of Chl *a* (a) and Chl *b* (b) overlaid with tri-exponential decays fits (lines) in MeOH (blue), THF (yellow), and  $\text{Et}_2\text{O}$  (green).

cyan lines superimposed onto the plot as shown in Figs. 3 and 4.

We will first present the  $\text{CLS}\omega_t$  results of Chl *a* in protic solvent MeOH, and solvents with lower polarity: THF and  $\text{Et}_2\text{O}$ . Fig. 5(a) shows the comparison of  $\text{CLS}\omega_t$  among different solvents for the first 200 ps for Chl *a* molecule. The  $\text{CLS}\omega_t$  data are then fitted with sum of exponential decays,  $\sum_i A_i \exp(-\tau_i/T_w)$  with amplitudes  $A_i$  and timescales  $\tau_i$ , and presented in Table 2. The results show a significant difference in term of the slope values between the highly polar hydrogen bonding solvent (MeOH) and the other aprotic nonpolar solvents. When the solvent polarity was varied from THF to  $\text{Et}_2\text{O}$ , it did not result in a significant change in the values of  $\text{CLS}\omega_t$ . This illustrates that the FFCF is most affected by the high polarity or hydrogen bonds in a solvent like MeOH and gives only a minimal effect to the change of polarity in nonpolar solvents, which is in agreement with existing results using the ellipticity method [24]. We performed similar studies on Chl *b* with the same set of solvents. The obtained results are plotted in Fig. 5(b) and were fitted to a sum of exponential decay with the fitted parameters presented in Table 3. The  $\text{CLS}\omega_t$  for Chl *b* shows a similar trend as Chl *a* between the highly polar hydrogen bonding solvent (MeOH) and the other aprotic nonpolar solvents. Likewise, when the solvent polarity was varied from THF to  $\text{Et}_2\text{O}$ , it did not result in a significant change in the values of  $\text{CLS}\omega_t$ .

The fastest time component  $\tau_1$  for Chl *a* and Chl *b* (Tables 2 and 3) in all the solvents ranges from 200 fs to 700 fs. We did not resolve with a higher level of precision, the sub-picosecond timescale  $\tau_1$  as we had not collected sufficient 2D spectra for  $T_w < 1$  ps. These values can be compared to related values obtained using time-resolved fluorescence (TRF) and three-pulse photon echo peak shift spectroscopy (3PEPS) solvation dynamics measurements. In TRF experiments using coumarin 153 molecule as a probe, the sub-picosecond fast timescale of the solvation dynamics in solvents MeOH and THF were measured to be in the range of 200 to 300 fs [16]. In a 3PEPS experiment using the dye molecule IR144 as a probe molecule performed by Joo *et al.*, a 50 to 70 fs timescale is observed for hydrogen bonding protic solvents as well as other aprotic solvents [17], while a similar 3PEPS experiment with DTTCl as the probe molecule measures a fast component of 260 fs [41]. Our measured values are consistent with these timescales in general, which are assigned to the inertial component of solvation that results from the libration motion of the solvent molecules. Further analysis of this sub-ps timescale can be done by using the method proposed by

Table 2

Fit parameters of tri-exponential decays of  $\text{CLS}\omega_t$  values for Chl *a*. The amplitudes are denoted in the square brackets.

| Solvent (Chl <i>a</i> ) | $\tau_1$ (ps) [ $A_1$ ] | $\tau_2$ (ps) [ $A_2$ ] | $\tau_3$ (ps) [ $A_3$ ] | $\tau_4$ [ $A_4$ ] |
|-------------------------|-------------------------|-------------------------|-------------------------|--------------------|
| MeOH                    | 0.27 [0.19]             | 4.63 [0.16]             | 42 [0.17]               | > 1 ns [0.03]      |
| THF                     | 0.45 [0.28]             | 2.44 [0.16]             | 32 [0.04]               | > 1 ns [0.01]      |
| $\text{Et}_2\text{O}$   | 0.57 [0.24]             | 4.51 [0.10]             | 59 [0.02]               | > 1 ns [0.02]      |

Table 3

Fit parameters of fitted tri-exponential decays of  $\text{CLS}\omega_t$  values for Chl *b*. The amplitudes are denoted in the squared brackets.

| Solvent (Chl <i>b</i> ) | $\tau_1$ (ps) [ $A_1$ ] | $\tau_2$ (ps) [ $A_2$ ] | $\tau_3$ (ps) [ $A_3$ ] | $\tau_4$ [ $A_4$ ] |
|-------------------------|-------------------------|-------------------------|-------------------------|--------------------|
| MeOH                    | 0.67 [0.12]             | 7.10 [0.18]             | 47 [0.22]               | > 1 ns [0.06]      |
| THF                     | 0.33 [0.24]             | 2.10 [0.23]             | 45 [0.04]               | > 1 ns [0.05]      |
| $\text{Et}_2\text{O}$   | 0.41 [0.35]             | 2.40 [0.21]             | 32 [0.06]               | > 1 ns [0.07]      |

Fayer *et al.* [30], to recover the motionally narrowed components of the spectra.

We next discuss the decay components in the picosecond timescales. Both Chl *a* and Chl *b* in all solvents have significant and comparable amplitudes,  $A_2$  in the  $\tau_2 = 2$  to 7 ps timescale range. However, for both the Chl *a* and Chl *b* in MeOH, there is also a prominent component,  $A_3$  of timescale  $\tau_3 \sim 40$  ps which is absent or insignificant in the nonpolar solvents THF and  $\text{Et}_2\text{O}$ . The  $A_3/A_2$  ratio for Chl *a* and Chl *b* in MeOH being 1.06 and 1.22, respectively compared to values between 0.17 and 0.28 for Chl *a* and Chl *b* in THF and  $\text{Et}_2\text{O}$ . Similar observations in TRF experiments are reported where the solvation dynamics of MeOH has similar relative amplitudes for the 3.2 ps and 15 ps timescale, while THF has only amplitude in the 1.5 ps timescale with no amplitude in the > 10 ps timescale [16]. The picosecond timescales in the solvation dynamics studies are attributed to the diffusive dynamics of the solvent molecules and involve the translation and rotation of the bulk solvent molecules [42]. The  $\sim 40$  ps timescale observed in both Chl *a* and Chl *b* in MeOH here may likely arise from the hydrogen-bond network that hinders and slows down the rearrangement of the MeOH molecules.

The slowest component  $\tau_4$  in the exponential decay correspond to a timescale that is larger than our maximum measured population time of  $T_w \sim 1$  ns. The amplitude for this timescale,  $A_4$  is consistently larger for Chl *b* in all three solvents as compared to Chl *a*. This component is a measure of the inhomogeneity of the system. As mentioned earlier, Chl *a* and Chl *b* have a very close resemblance in terms of the structure. The distinction comes from the side group substituent in which the methyl group at position 7 in Chl *a* is replaced with a formyl group (Fig. 1). There have been numerous experiments performed, trying to identify the role of this formyl group in the hydrogen bond interaction with the protein residual in the photosynthetic membrane [10,13,43–45]. In the structure of Chls, the carbonylic oxygen in the porphyrin ring is the most nucleophilic center serving as the hydrogen-bond recipient [46]. In protic solvents such as MeOH, it forms a hydrogen bond with the  $-\text{OH}$  group of the solvent ( $\text{C}=\text{O}\cdots\text{H}-\text{O}$ ). Therefore, two of the potential binding sites ( $13^1$  and  $13^3$ ) that can form this type of hydrogen bond interaction can be found in Chl *a* tetrapyrrole ring [11], whereas an extra hydrogen bond site exists in the case of Chl *b* with the formyl group ( $-\text{CH}=\text{O}$ ) at position 7 [13]. This extra hydrogen bond with MeOH will limit the faster fluctuation of the porphyrin ring and thus the frequency fluctuation of Chl *b* electronic transition and explain the

higher inhomogeneity observed in Chl *b* than Chl *a*. For both Chl *a* and Chl *b*, the amplitudes for this slow component  $A_4$ , are small compared to the overall sum of the other components. This suggests that the major contribution to the broader linear spectrum linewidth of Chl in MeOH compared to in THF and Et<sub>2</sub>O (See Table 1) is not due to inhomogeneity.

In another study, Hooper *et al.* have proposed that modification at the periphery of the Chl molecules influences the coordination chemistry of the central Mg atom and this effect plays an essential role in the interaction of Chl *b* with LHC apoprotein and thus in the assembly of LHCs [45,47,48]. For the case of Chl *b* in THF and Et<sub>2</sub>O, even though hydrogen bond interaction is not formed between the formyl groups and the solvent molecules, the 7-formyl group in Chl *b* is reported to have an effect to the strength of interaction between ligand and the metal center of the porphyrin (Mg). As proposed by Hooper and Eggink, the metal center in Chl *b* expresses stronger positive charge than Chl *a* [45,47]. The reason is that the additional formyl group in Chl *b* pulls the electron density from the pyrrole nitrogen towards the substituent group and thus makes the Mg atom less shielded by the electron cloud than in Chl *a*. As a result, the negative end of a fixed dipole (solvent) will have a stronger interaction with the positively charged group of Mg<sup>2+</sup> ion. This argument has been supported experimentally by measurement of the equilibrium constant of various tetrapyrrole derivatives with specific ligands done by Tamiaki *et al.* [49]. Although it is not straightforward to relate this measurement with quantitative values of the FFCF, it is likely that the measured FFCF can be associated with the strength of ligation between solvent molecules and central Mg<sup>2+</sup> ion as speculated earlier by Moca *et al.* [24]. For Chl *b*, this can increase the probability of the solvent-Mg<sup>2+</sup> ion interaction to persist beyond 1 ns. There may be a distribution in the configurations of this solvent-Mg<sup>2+</sup> ion interaction thus giving rise to higher  $A_4$  values of the fit to the CLS $\omega_t$  values as observed for Chl *b*.

#### 4. Conclusion

The spectral diffusion process of the  $Q_y$  transition of Chl *a* and Chl *b* has been studied using 2DES. Although there have been several 2DES spectral diffusion studies performed on Chl *a* [22,24], this is the first report of similar studies on Chl *b*. By utilizing the CLS $\omega_t$  method to extract the normalized FFCF from the 2D spectra, we have compared the spectral diffusion dynamics of Chl *a* and Chl *b* in protic solvent MeOH and nonpolar aprotic solvents THF and Et<sub>2</sub>O. A clear result is that for both Chl *a* and Chl *b*, there is a ~40 ps component in protic solvent methanol, that is absent in aprotic solvents, tetrahydrofuran and diethyl ether. This component may be attributed to the solvent's hydrogen-bond network dynamics. With such information, together with comparisons with measured spectral diffusion dynamics of Chls in more complex environment such as the protein environment in photosynthetic complexes, scientists will be able to have a better idea of the protein environment's interaction with these Chls. We also observed differences in values in CLS $\omega_t$  of the two Chls despite their close resemblance in terms of structure. Higher contributions to inhomogeneous broadening as observed for Chl *b* were attributed to the effect of the hydrogen bond interaction formed with the additional formyl group and the stronger electrostatic interaction between the metal center of porphyrin (Mg<sup>2+</sup>) and ligand (solvent) resulting from the electron withdrawing effect from the formyl group.

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